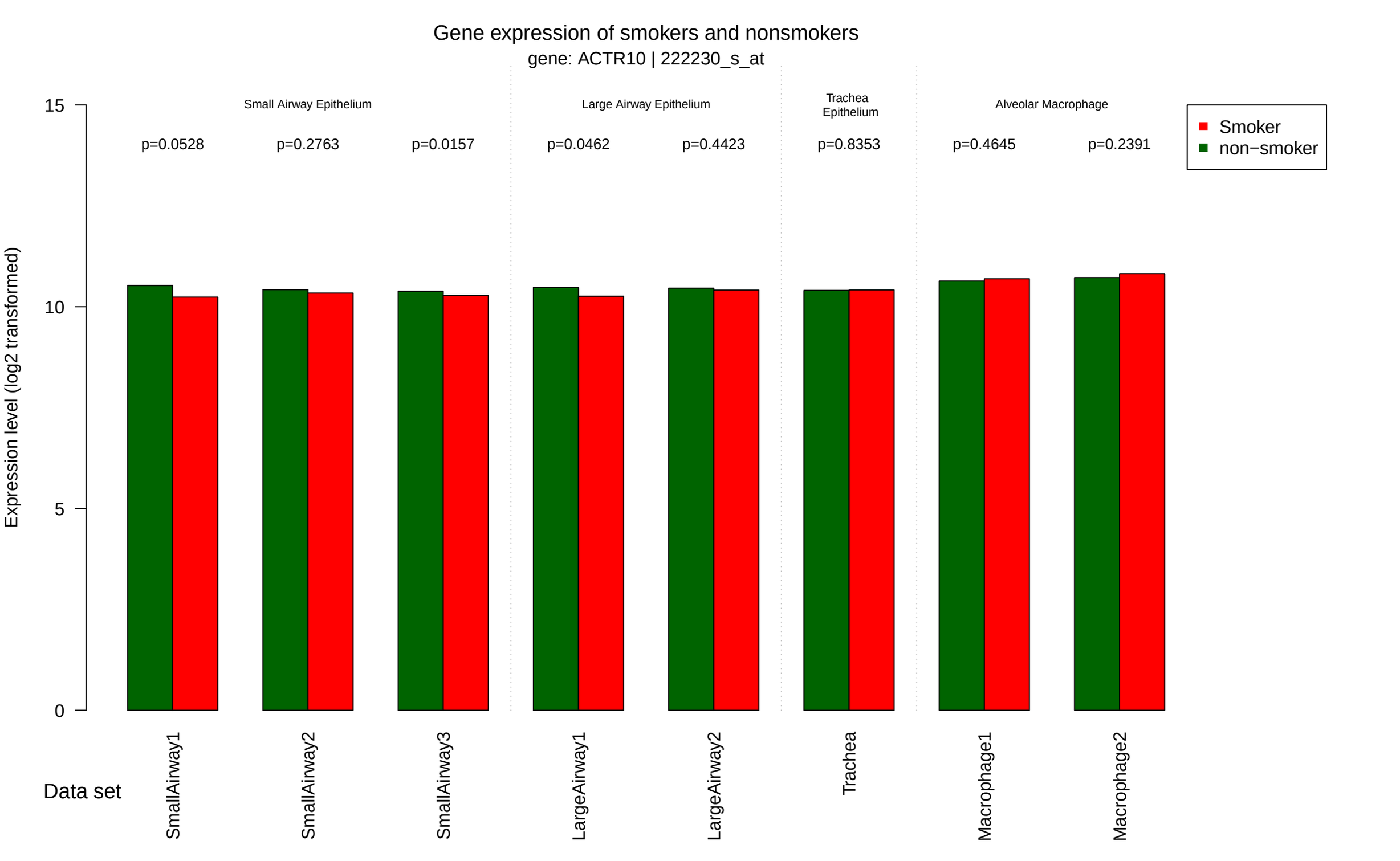
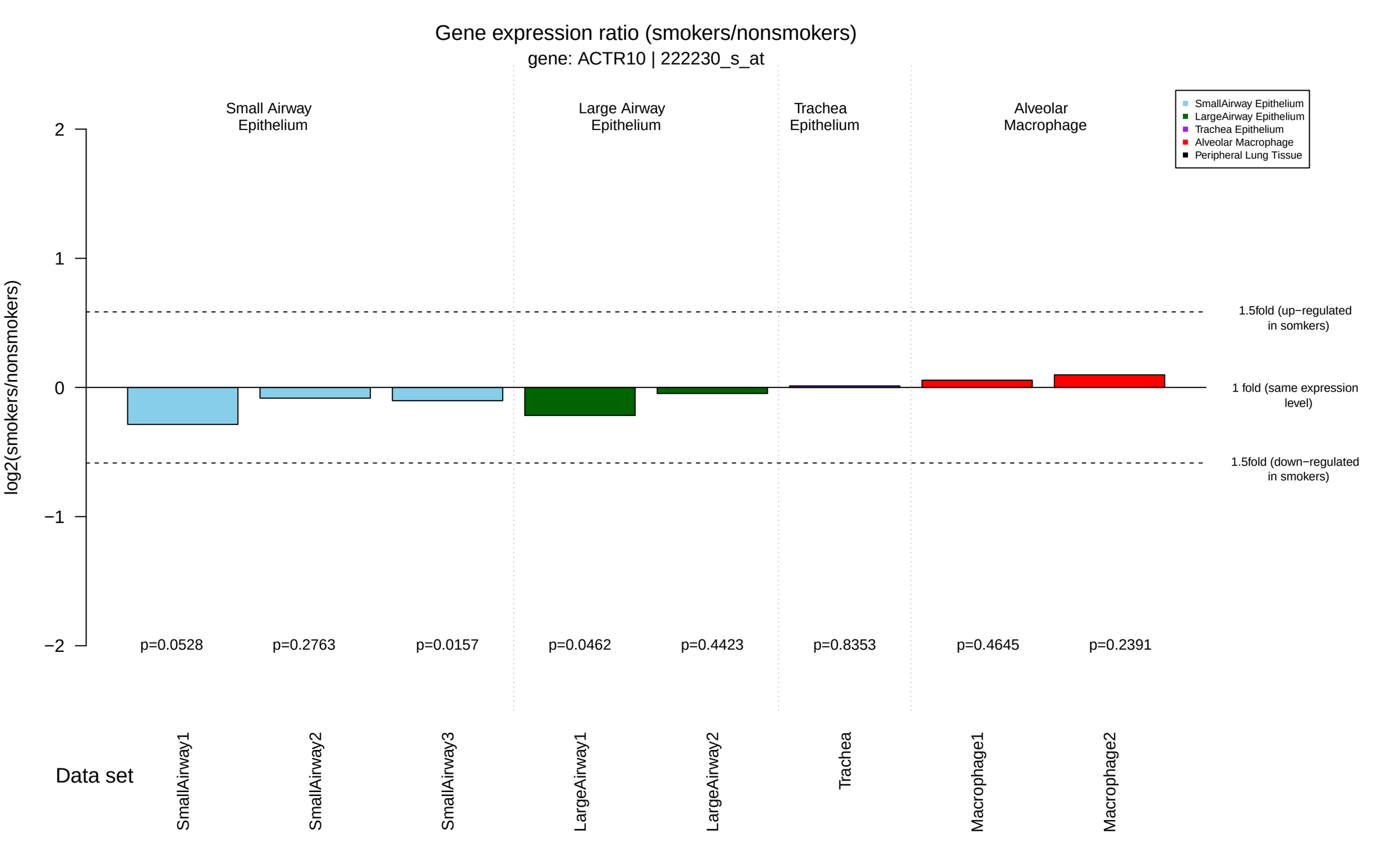
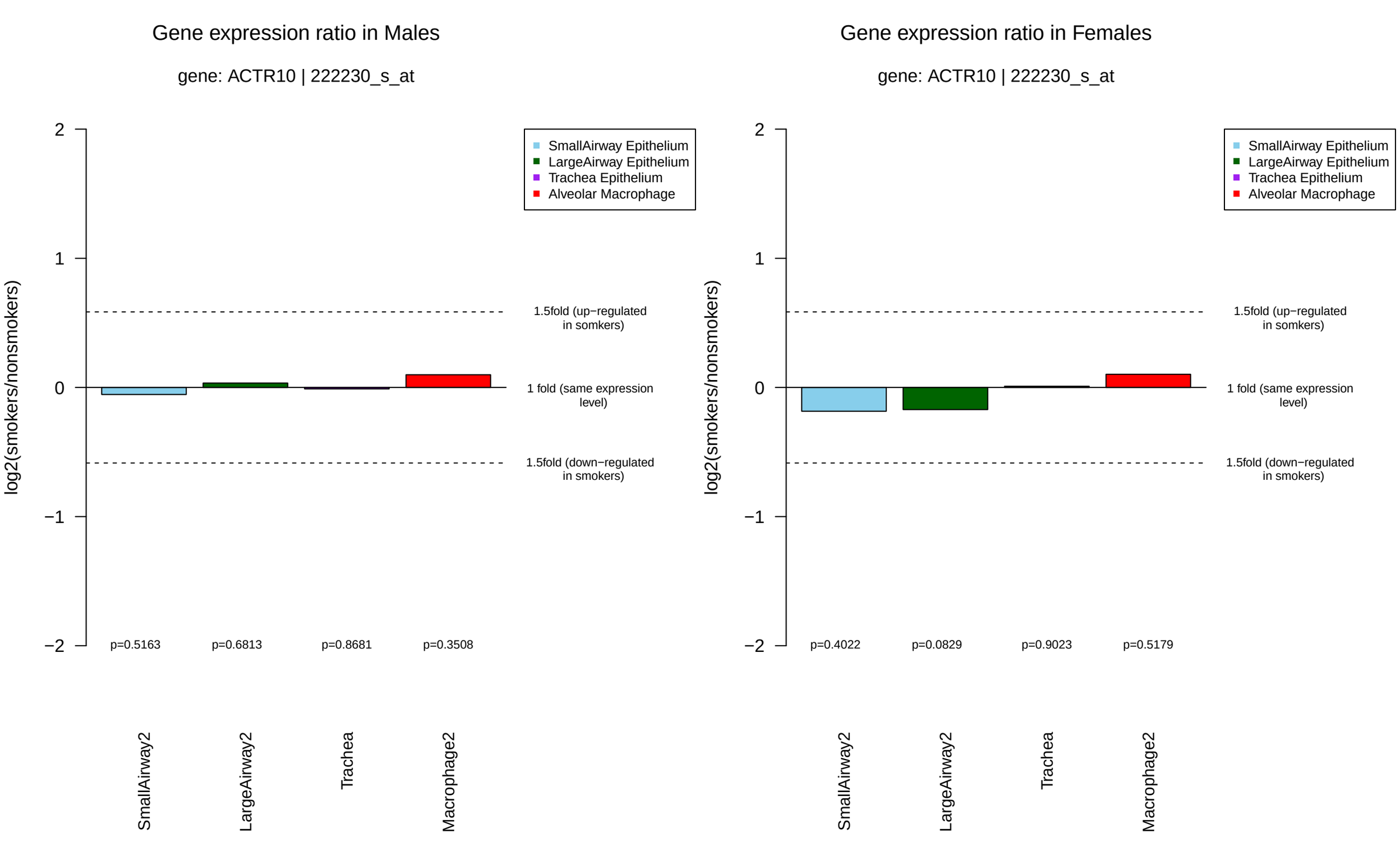
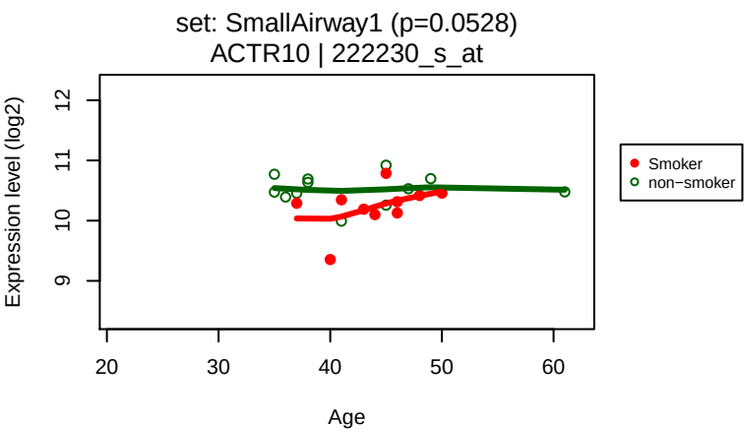
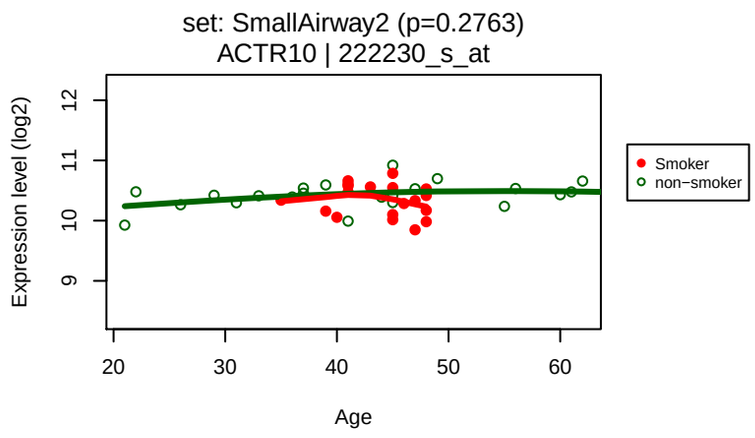
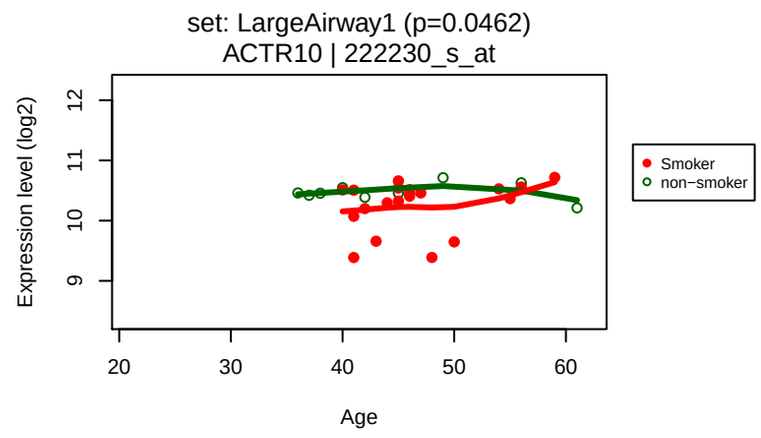

Gene expression of smokers and nonsmokers
gene: ACTR10 | 222230_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0528
p=0.2763
p=0.0157
p=0.0462
p=0.4423
p=0.8353
p=0.4645
p=0.2391
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

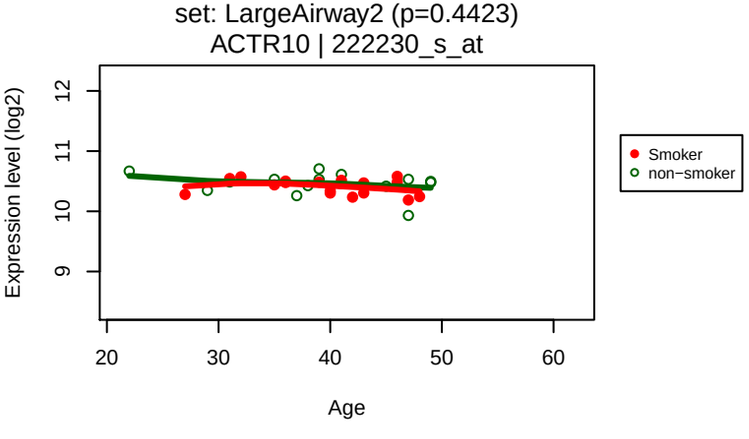

Gene expression ratio (smokers/nonsmokers)
gene: ACTR10 | 222230_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0528
p=0.2763
p=0.0157
p=0.0462
p=0.4423
p=0.8353
p=0.4645
p=0.2391
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

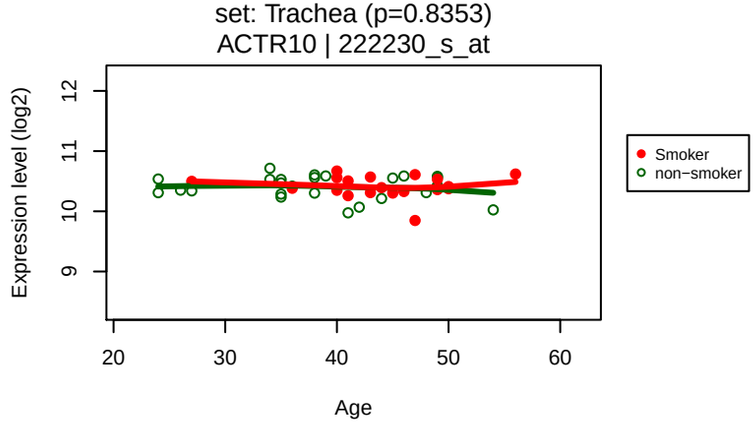

Gene expression ratio in Males
gene: ACTR10 | 222230_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5163
p=0.6813
p=0.8681
p=0.3508
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACTR10 | 222230_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4022
p=0.0829
p=0.9023
p=0.5179
SmallAirway2
LargeAirway2
Trachea
Macrophage2

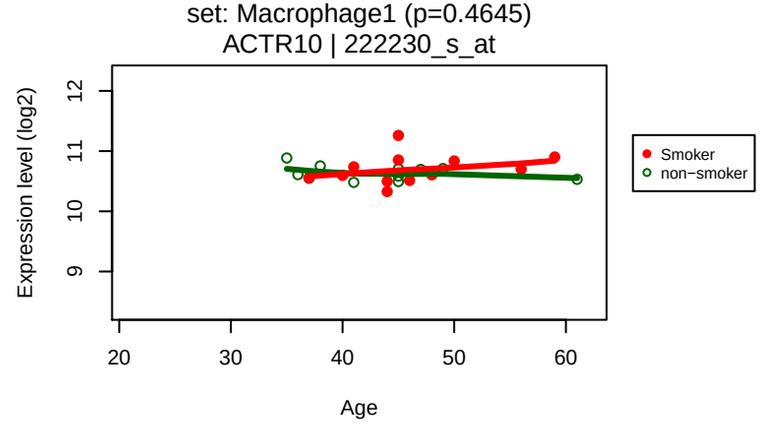

set: SmallAirway1 (p=0.0528)
ACTR10 | 222230_s_at
Expression level (log2)
Age
Smoker
non−smoker

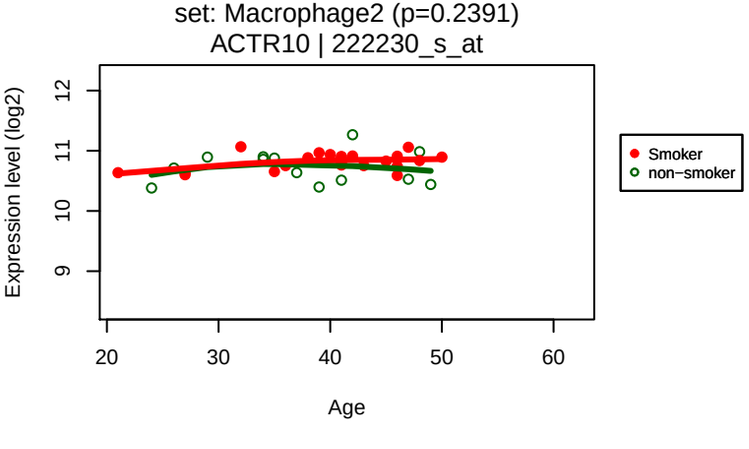

set: SmallAirway2 (p=0.2763)
ACTR10 | 222230_s_at
Expression level (log2)
Age
Smoker
non−smoker

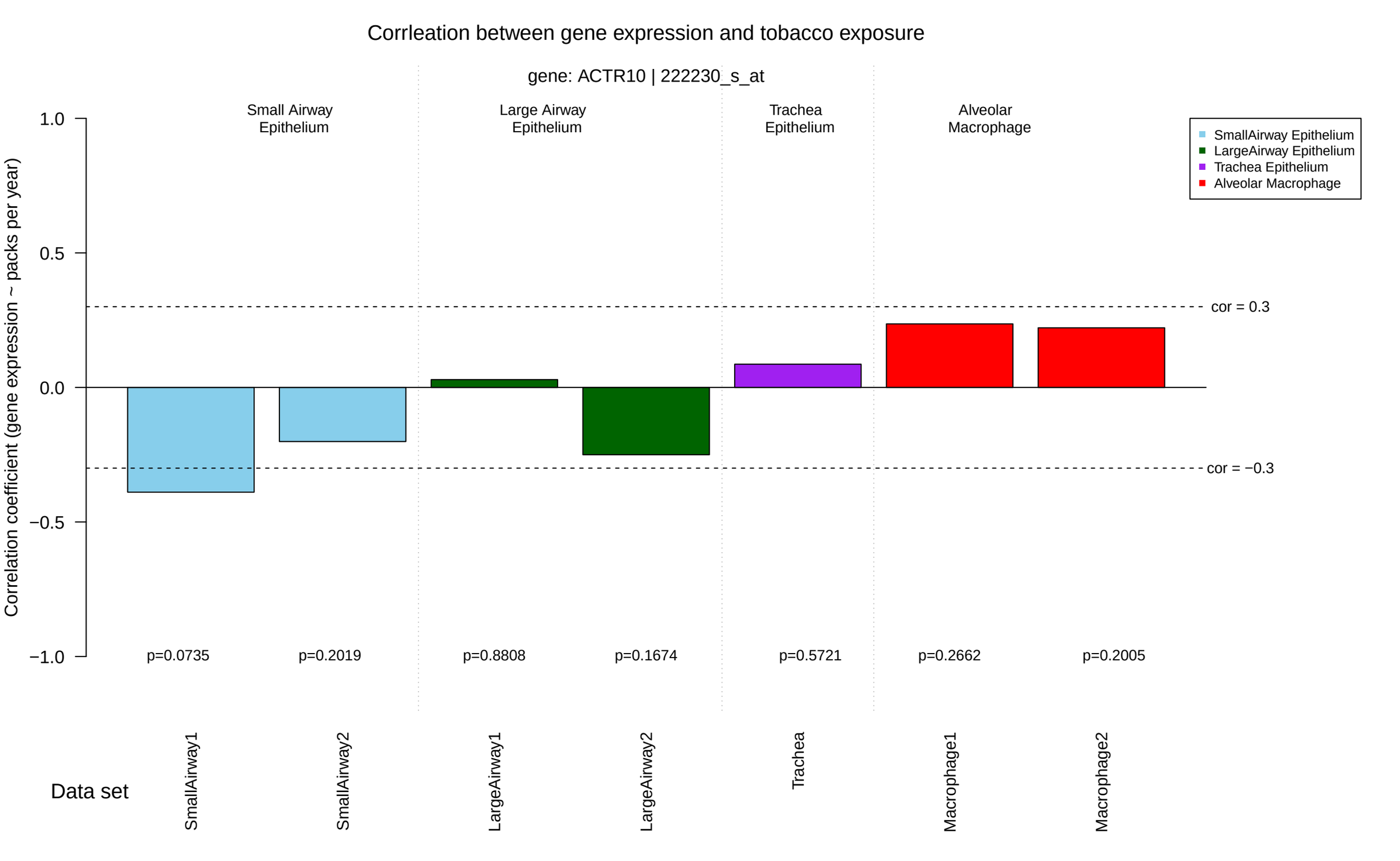

set: LargeAirway1 (p=0.0462)
ACTR10 | 222230_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4423)
ACTR10 | 222230_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8353)
ACTR10 | 222230_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4645)
ACTR10 | 222230_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2391)
ACTR10 | 222230_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACTR10 | 222230_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0735
p=0.2019
p=0.8808
p=0.1674
p=0.5721
p=0.2662
p=0.2005
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2